22/1/26

EXP 1

AZURE DEVOPS ENVIRONMENTAL SETUP

* **Aim:**

To set up and access the Azure DevOps environment by creating an Organization through the Azure Portal.

* **INSTALLATION:**

1. Open your web browser and go to the Azure Website:
https://azure.microsoft.com/en-us/get-started azure-portal.
Sign in using your Microsoft account credentials.
If you don't have a Microsoft account, you can create one here:
https://signup.live.com/?lic=1

2. Azure home page
3. Open DevOps environment in the Azure platform by typing Azure DevOps Organizations in the search bar.

4. Click on the My Azure DevOps Organization link and create an Organization and you should be taken to the Azure DevOps Organization Home page.

RESULT:

Successfully accessed the Azure DevOps environment and created a new Organization through the Azure portal.

EXPERIMENT 2

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AZURE DEVOPS PROJECT SETUP AND USER STORY

Create Epic, Features, User stories, Task

Aim:
To set up an Azure Devops project for efficient collaboration and agile work management.

1. Create An Azure Account
2. Create the First Project in Your Organization

a. After the Organization is set up, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more.

b. On the Organization's Home page, Click on the New Project button.

c. Enter the project name, description and Visibility options:

Name: Choose a name for the project (e.g., LMS).

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be private (accessible only to those invited) or Public (accessible to anyone).

d. Once you've filled out the details, click Create to set up your first project.

3. Once logged in, ensure you are in the Correct Organization. If you're part of multiple Organizations, you can switch between them from the top left corner (next to your user profile).
Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

4. Project dashboard

5. To manage user stories:

a. From the left-hand navigation menu, click on Boards. This will take you to the main Boards page, where you can manage work items, backlogs and sprints.

b. On the work items page, you'll see the option to Add a work item at the top.

Alternatively, you can find a + button or Add New Work Item depending on the view you're in. From the Add a work item drop down, select User story. This will open a form to enter details for the new User story.

RESULT:

Successfully created an Azure DevOps project with user story management and agile workflow set up.

EXPERIMENT – 3

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SETTING UP EPICS, FEATURES AND USER STORIES FOR PROJECT PLANNING

AIM:

To learn about how to create epics, user story, features / backlogs for your assigned project.

Create Epic, Features, User Stories, Task.

1. Fill in Epics
2. Fill in Features
3. Fill in User story Details

RESULT:

Thus, the creation of epics, Features, User story and task has been created successfully.

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EXPERIMENT - 4

SPRINT PLANNING

AIM:

To assign user story to specific sprint for the Music playlist Batch Creator Project.

Sprint Planning

Sprint 1

Sprint 2

Sprint 3

Sprint 4

RESULT:

The Sprints are created for the Music playlist Batch Creator Project.

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EXPERIMENT-5

POKER ESTIMATION

AIM:

Create Poker Estimation for the User stories – Music Playlist Batch Creator Project

Poker Estimation:

* In azure microsoft account click your main ongoing project
* Click New and create a user story and add story points
* Each members in the team should add their story points in planning column.

Ex: member 1 : 3

member 2 : 5

member 3 : 8

(tester) QA : 8

Finally every team members go with story point 5, according poker estimation.

RESULT:

The Estimation / Story Points is created for the project using Poker Estimation.

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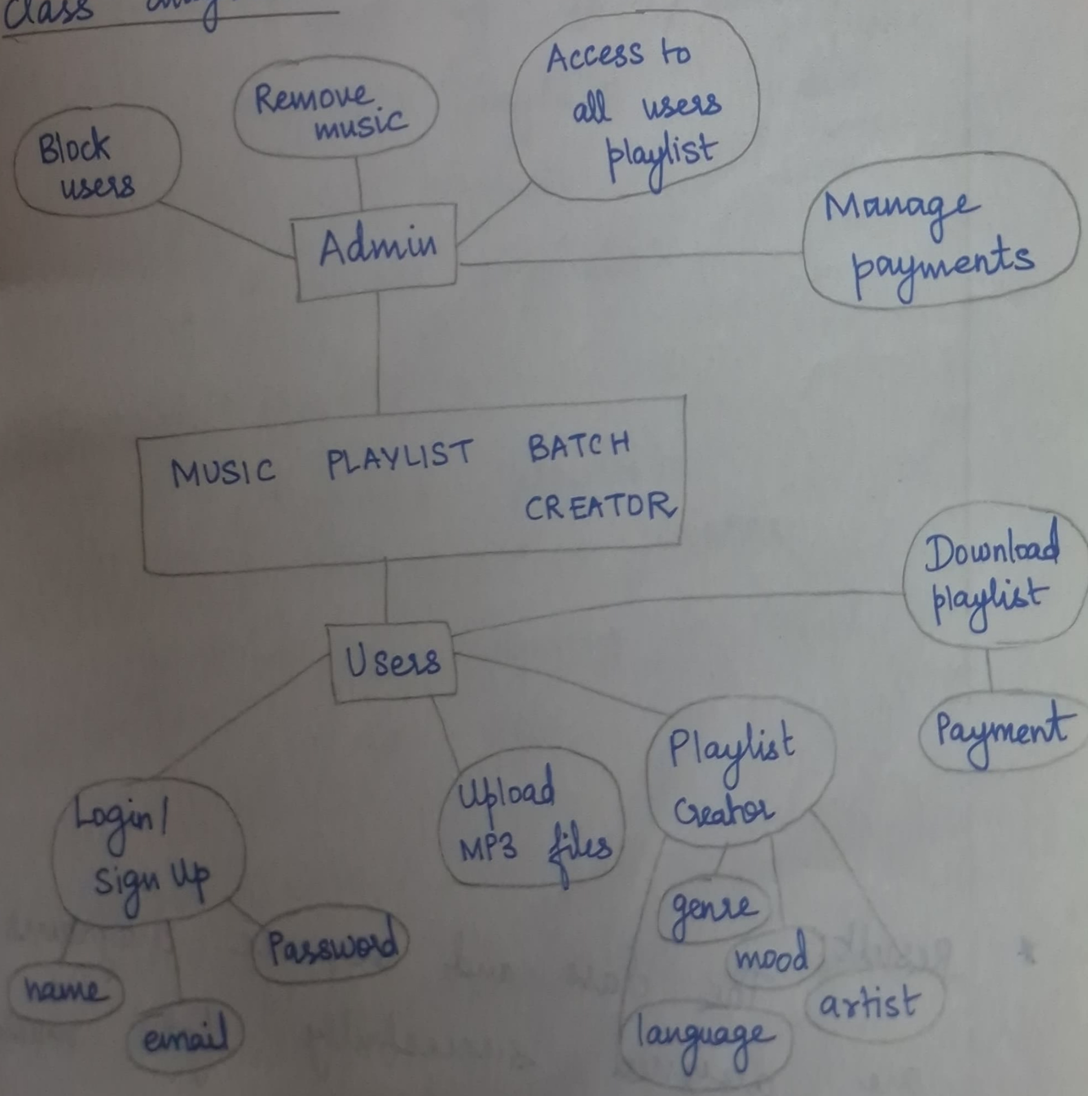
EXPERIMENT - 6

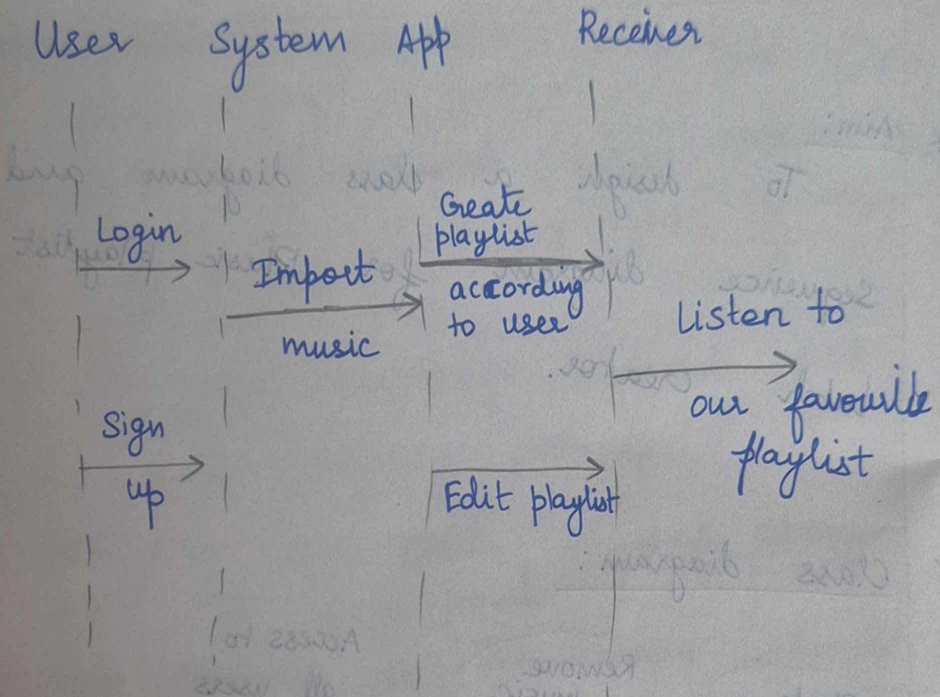
DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

* Aim:
To design a class diagram and sequence diagram for Music playlist batch creator.

* Class diagram:



* Sequence diagram



*** Result:** The class and sequence diagrams are designed succesfully for music playlist creator.

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EXPERIMENT - 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

* AIM :- To design an architectural diagram and ER diagram for the Music playlist batch creator.

* User Interface (UI):

It is used by users/admin to:

→ create playlists

→ edit playlists

→ delete playlists

* Application layer:

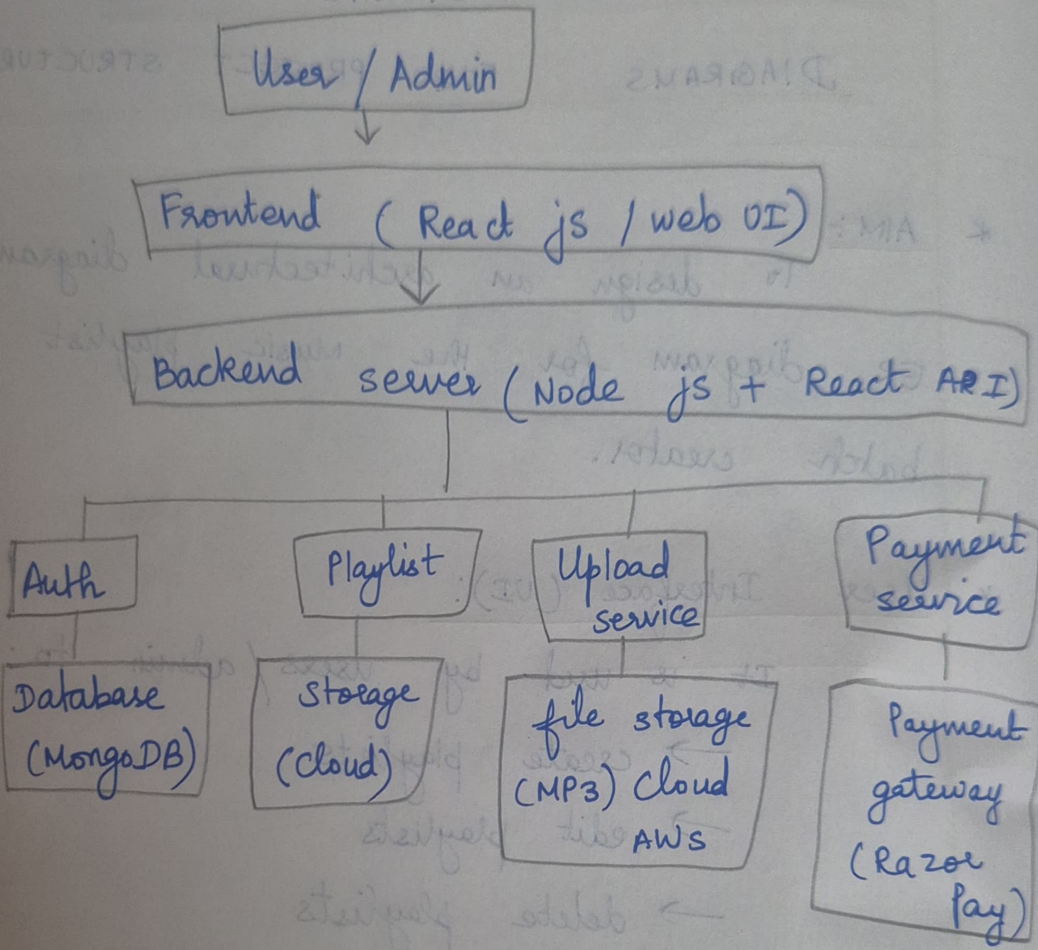
→ handles playlists according user wishes

→ creates according to mood, artists, genre

* Database layer:

→ Stores music which is only imported

* Diagram :-



* Result :-

The architechural diagram ER diagram are designed successfully for the music playlist creator

EXPERIMENT - 8

Testing (Test Plans & Test cases)

* AIM:
To design test plans and test cases for the Music Playlist batch creator project.

* Description:
Test plans were created for key features like login, playlist creation, editing and metadata. Both happy path and error scenarios were tested to ensure reliability.

Sample Test cases:

- Login Success: User enters valid credentials → redirected to dashboard
- Login Failure: Wrong Password → error message shown
- Create Playlist: Playlist created successfully
- Empty Playlist Name: Error displayed
- View Playlist: All playlists shown
- Offline Mode: Playlist loading fails with error

* <u>Result</u>:

Test cases were successfully created and executed for major user stories ensuring proper functionality.

EXPERIMENT - 9

Load Testing and Pipelines

* **AIM:**

To perform load testing and implement CI/CD pipeline for the project.

* **Description:**

- Load testing was done using Azure to check performance under high traffic
- CI/CD pipeline was created using Azure DevOps connected with GitHub.

Pipeline Tasks:

- Code checkout
- Python environment setup
- Install dependencies
- Run test script

Outcome:

- Application handled multiple requests efficiently
- Pipeline automated build and testing process

Result:

Load testing and pipelines setup were completed successfully for performance & automation.

EXPERIMENT-10

GitHub Project Structure & Naming

* **AIM:**

To organize the project structure & maintain proper naming conventions

* **Project Structure Example:**

- /project → main application code
- /templates → HTML files
- /static → CSS, JS
- /tests → test cases
- requirements.txt → dependencies
- app.py → main file

* **Naming conventions:**

- Files: lowercase with underscores
  (playlist_manager.py)
- Variables: meaningful names
  (user_playlist)

- Functions : action-based
  (create_playlist ())

* **Result** :

Project structure was organized clearly, improving reliability & maintability.